

Total No. of Questions : 10]

SEAT No. :

P2903

[6004]-801

[Total No. of Pages : 2

**B.E. (Computer Engineering)
INFORMATION AND CYBER SECURITY
(2015 Pattern) (Semester - II) (410251)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No. 1 or Q. No. 2, Q.No. 3 or Q.No. 4, Q.No. 5 or Q.No. 6, Q.No.7 or Q.No.8, Q.No.9 or Q.No.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain operational model of Security with neat diagram. [5]
b) Use transposition cipher to encrypt message "WE ARE THE BE" use key "HEAVEN". [5]

OR

- Q2)** a) Using Hill cipher encrypt plain text "COE" Key is "ANOTHERBZ". [5]
b) Compare between symmetric key encryption and asymmetric key encryption. [5]
- Q3)** a) Discuss Diffie-Hellman Key exchange algorithm with the help of suitable example. [5]
b) Explain operation of DES algorithm in detail. [5]

OR

- Q4)** a) What is authentication? Explain various methods of authentication. [5]
b) What is block cipher? Explain counter mode of block cipher. [5]
- Q5)** a) Discuss the working of IPsec. What are the benefits of IPsec? [7]
b) What is VPN? Explain types of VPN. [7]
c) Write short note on E-mail Security. [4]

OR

P.T.O.

- Q6)** a) Explain working of PGP in detail. What is backdoors and Key Escrow in PGP? [7]
b) Explain Secure Electronic Transaction in detail. [7]
c) Differentiate between IP-V4 and IP V6. [4]

- Q7)** a) What are the various types of firewall? Discuss packet filtering firewall. [6]
b) Explain any two password management practices. [6]
c) Explain need and challenges of intrusion detection systems. [4]

OR

- Q8)** a) What is intrusion detection system? Enlist and explain different types of IDS. [6]
b) Explain access control security services. [6]
c) Explain characteristics of Firewall. [4]

- Q9)** a) Explain personally identifiable information PII. Describe PII impact levels with examples. [8]
b) What is cyber stalking? How to identify and detect cyber stalking. [8]

OR

- Q10)** a) What is Cyber crime? Explain with suitable example. Discuss various types of cyber crime. [8]
b) Discuss PII confidentiality safeguards [8]

